



TITLE PAGE

- **Problem Statement ID** – SIH26060
- **Problem Statement Title** – Digital Platform for efficient remote management of Indian Antarctic Research Stations
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – <your team ID>
- **Team Name** – Team Atrangi



ANTARVIK

A unified digital twin for Maitri & Bharati stations

IDEA TITLE

PROPOSED SOLUTION • IDEA / SOLUTION / PROTOTYPE

ANTARVIK

ANTARctica + VIVEK — Sanskrit for discernment

An operator-facing digital twin of Maitri & Bharati stations

Four domains — infrastructure, energy, logistics, environment — fused in ONE coupled model, not four dashboards.

THE COUPLING THAT MAKES IT A TWIN, NOT A DASHBOARD — LIVE DEMO CHAIN



1 WHAT IT IS

- Federated, edge-first twin of both stations — one registry, more stations plug in
- Four subsystem twins joined by a cross-domain coupling engine (physics + ML)
- Five layers: asset → telemetry → model → analytics → decision
- Operator console: alerts, work orders, resupply plans, full-season replay
- Synthetic telemetry built from documented station specs, with adapters for real NCPOR feeds

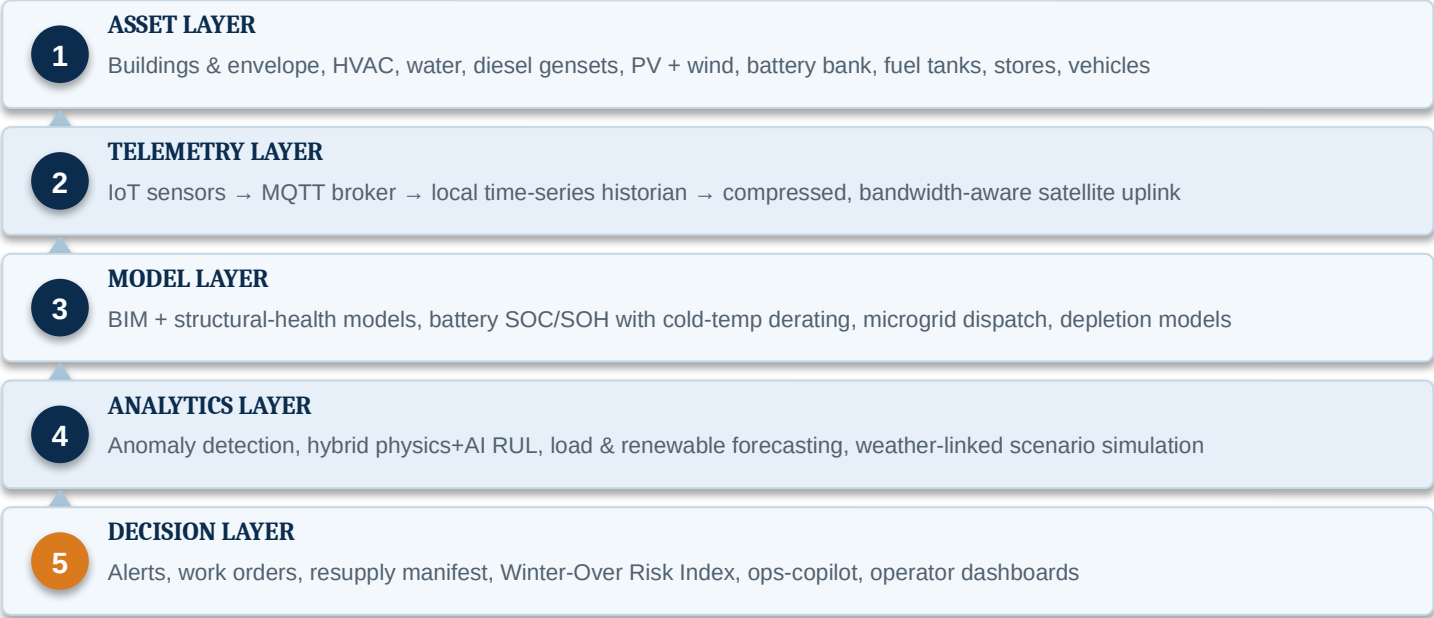
2 HOW IT SOLVES THE PROBLEM

- NCPOR Goa has zero live view of fuel, power, stores or building health, 4,000 km away
- Predicts winter shortfalls months ahead of the single annual resupply window
- Keeps working through satellite blackouts: Connected → Degraded → Recovery
- Unifies NCPOR's existing silos (Bharati-TIMES, itinerary tools) into one layer
- Sensing in Antarctica is already mature — the unsolved problem is integration

3 INNOVATION & UNIQUENESS

- **Born-Digital Maitri II:** the ₹2,000 cr DPR is being written now — the twin ships with the station, sensors specified pre-build
- **Winter-Over Risk Index:** fuel margin + food-days + medical stores + power redundancy as one forecast score
- **Sea-ice-aware resupply router:** ice delays the vessel → auto-recompute winter margins + airlift shortlist
- **Degraded-mode governance:** the twin honestly drops to a shadow, then forensic replay on reconnect
- **LLM ops-copilot:** explains alerts, drafts work orders, answers “what if a genset fails in winter?”

METHODOLOGY • FEDERATED 5-LAYER ARCHITECTURE



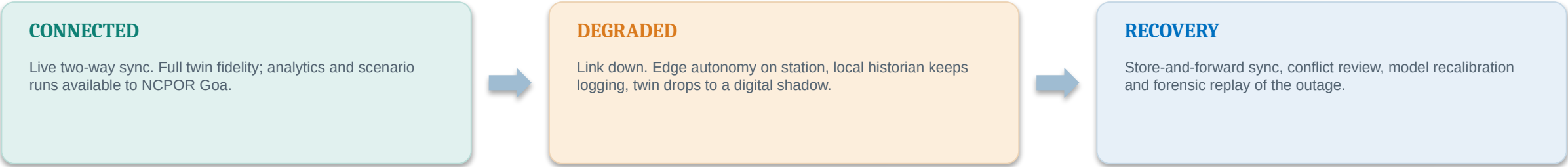
TECHNOLOGIES TO BE USED

- **3D twin & UI:** Three.js · CesiumJS · React
- **Coupling engine:** Python · FastAPI · Docker · REST/WebSocket
- **Data plane:** MQTT (Mosquitto) · TimescaleDB / InfluxDB
- **Forecasting:** Prophet · sktime · LSTM
- **Anomaly & RUL:** Isolation Forest · LSTM-Autoencoder · physics-informed hybrid
- **Simulation:** microgrid dispatch, HOMER-style calibration
- **Ops-copilot:** LLM + RAG over station logs, manuals and twin state
- **Security:** mTLS device onboarding · RBAC · signed local audit logs

WHERE THE DATA COMES FROM

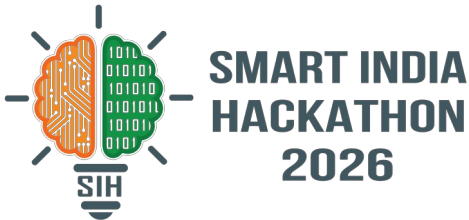
No dataset ships with this PS. We generate physically-plausible telemetry from the publicly documented station layout, crew size and resupply cycle, calibrated against published Antarctic energy studies — and expose an adapter layer so real NCPOR feeds replace the simulators with no re-architecture.

PROCESS UNDER A BROKEN SATELLITE LINK — THREE OPERATING MODES





FEASIBILITY AND VIABILITY



FEASIBILITY OF THE IDEA

TECHNICALLY BUILDABLE

- Fully open-source stack, no exotic hardware — one edge box on station, one server at NCPOR
- Hackathon MVP: coupling engine + two live scenarios + degraded-mode failover

CALIBRATED, NOT GUESSED

- Simulators tuned to peer-reviewed Antarctic station parameters, so the numbers survive scrutiny

170 kW
station load

124,000 gal/yr
diesel baseline

96%
fuel saving, PV+wind

3×240 kVA
genset configuration

40%
exhaust heat recovered

+1.8%
sim vs measured

Sources: NREL REopt South Pole study · Comandante Ferraz (Brazilian Antarctic Station) energy study

INSTITUTIONALLY VIABLE

- Adapter layer ingests real NCPOR feeds (Bharati-TIMES, met, station WiFi) with no re-architecture
- The Maitri II project report is being drafted now — sensor spec can go in before construction
- Cheaper and more sovereign than Siemens/Bentley suites, which ship no polar logistics couplings

POTENTIAL CHALLENGES, RISKS → OUR STRATEGY

- No dataset is released by NCPOR** → physics-calibrated synthetic generators, labelled honestly; live-feed adapters ready from day one
- Iridium-class link: minutes of bandwidth, hours of outage** → edge-first design, compressed delta sync, Degraded mode keeps the station autonomous
- Sparse failure history breaks pure-ML prognostics** → hybrid physics + AI RUL, so physics carries the model where data is thin
- The twin drifts out of step with the real station** → time-discrepancy monitoring (Fraseri 2023) and recalibration on every Recovery sync
- Automation in a life-critical setting must be earned** → human-in-the-loop throughout; closed-loop control only in Phase 4, bounded and supervised
- Cyber exposure and data sovereignty** → on-prem at NCPOR, signed device onboarding, RBAC, local audit logs that survive an outage

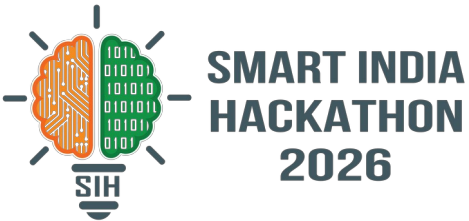
PHASED ROADMAP — STAGED ADOPTION, NOT A BIG-BANG ROLLOUT



SIH prototype scope covers Phase 1–3 end-to-end on synthetic telemetry; Phase 4 is deliberately deferred until the twin has earned trust on station.



IMPACT AND BENEFITS



₹2,000 cr

Maitri II rebuild approved Oct 2025 — a station the twin can be born with

96%

diesel displacement shown achievable at a polar station with PV + wind + battery

1

resupply window per year — one missed item costs a whole wintering season

4,000 km

between NCPOR Goa and two stations it currently manages with no live view

BENEFITS OF THE SOLUTION

OPERATIONAL

- Live view of fuel, power, stores and structural health across the whole station
- Failures forecast rather than discovered mid-winter
- Operations continue through satellite blackouts
- Winter-Over Risk Index gives the station leader one number to act on

ECONOMIC

- Lower diesel spend and fewer emergency airlifts
- Optimal cargo manifest means less dead weight on the annual voyage
- Condition-based maintenance extends the life of imported assets
- One framework amortised across stations, and later polar vessels

ENVIRONMENTAL

- Renewable utilisation up, diesel litres and CO₂ down
- Auto-generated Antarctic Treaty and CCAMLR compliance reporting
- Spill, waste and emissions risk tracked continuously
- Supports net-zero polar operations, the direction BAS has already set

STRATEGIC & NATIONAL

- UK (BAS PolarDT) and EU (ESA 4DAntarctica) have polar twins; India has none
- Maitri II handed over BIM-native instead of retrofitted years later
- Replicates to Himadri in the Arctic and to polar research vessels
- Builds indigenous polar-operations capability inside MoES and NCPOR

WHO BENEFITS

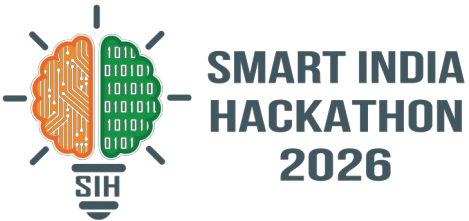
- NCPOR and MoES programme managers in Goa
- Station leader, engineers and the wintering crew
- Logistics and resupply planners for the annual voyage
- Maitri II DPR architects and MoES decision-makers

KPIs THE PLATFORM MOVES

- Unplanned downtime down · asset availability up
- Renewable utilisation up · diesel litres and CO₂ down
- Forecast accuracy on load, weather and stock depletion
- Inventory visibility, lead time, and data availability under a degraded link



RESEARCH AND REFERENCES



PRIOR ART — AND THE GAP WE FILL

BAS PolarDT — UK

Twins of Antarctic stations and ships, IceNet sea-ice forecasting, PolarRoute ship routing.

Not deployed as an integrated station-operations platform.

ESA 4DAntarctica — EU

€1.35 m Digital Twin of Antarctica for ice-sheet hydrology, under Digital Twin Earth.

Continent-scale science, not station operations.

NavVis / RV Polarstern

Walkable 3D digital twin of a polar research vessel used for logistics planning.

Geometry and logistics only; no energy or condition coupling.

NCPOR in-house tools

Bharati-TIMES life-support issue tracking, station WiFi, expedition itinerary tools.

Useful but siloed — no cross-domain twin sits above them.

No published work unifies meteorology, imagery, power and facility telemetry in a single Antarctic station twin. Sensing is mature — integration is the open problem, and that is exactly where ANTARVIK sits.

DIGITAL TWINS & POLAR OPERATIONS

- **Spasova, D. (2023).** Digital Twin and Polar Digital Space in Antarctica. SPIE Proceedings.
- **British Antarctic Survey** — PolarDT; IceNet six-month sea-ice forecasting; PolarRoute carbon-saving routing. bas.ac.uk
- **ESA 4DAntarctica** — Digital Twin Earth programme, University of Edinburgh. 4dantarctica.org
- **NavVis / AWI** — digital twin of RV Polarstern for logistics planning.
- **Fraseri, M. et al. (2023).** Time discrepancy between a digital twin and its physical counterpart.
- **Williams et al. (2026).** Digital twin + LLM for cross-domain predictive maintenance. Computers & Industrial Engineering.

ANTARCTIC ENERGY, LOGISTICS & STATION DATA

- **NREL REopt** — South Pole Station: 170 kW load, 124,000 gal/yr diesel; PV + wind + battery gives 96% fuel saving, 2–4 yr payback (arXiv).
- **Comandante Ferraz, Brazilian Antarctic Station** — 3×240 kVA gensets (duty/redundancy/emergency), CHP recovering 40% of exhaust heat; HOMER matched measured consumption within +1.8%.
- **Zhang et al. (2021).** Economic dispatch for an Antarctic station microgrid.
- **Lin et al. (2021).** Remote monitoring platform for an Antarctic greenhouse — direct precedent.
- **Sensors (2021)** — bandwidth-constrained Antarctic telemetry sensor networks.
- **MoES / PIB** — Maitri II approved 15 Oct 2025, ₹2,000 cr, solar + wind hybrid, completion ~2032; NCPOR Bharati-TIMES.

STANDARDS & FRAMEWORKS WE BUILD AGAINST: ISO 23247 digital-twin reference architecture · IEC 62443 for OT/edge security · Antarctic Treaty System environmental reporting · CCAMLR · MoES / NCPOR expedition SOPs